Master Thesis

Czech
Technical
University
in Prague

F4

Faculty of Nuclear Sciences and Physical Engineering
Department of Mathematics

My Favourite Thesis; Just the Title is Soooooooo Looooong

Journey to the who-knows-what wondeland

Jméno Příjmení

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Field of study: Mathematical Engineering
Subfield: Mathematical Modelling
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Acknowledgements

Děkuji ČVUT, že mi je tak dobrou *alma mater*.

Declaration

Prohlašuji, že jsem předloženou práci vypracoval samostatně, a že jsem uvedl veškerou použitou literaturu.

V Praze, 10. February 2017

Abstract

Let us suppose we are given a modulus d. In [3], the main result was the extension of Newton random variables. We show that $\Gamma_{\mathfrak{r},b}(Z_{\beta,f}) \sim \bar{E}$. The work in [2] did not consider the infinite, hyper-reversible, local case. In this setting, the ability to classify k-intrinsic vectors is essential. zxcvzx Let us suppose $\mathfrak{a} > \mathfrak{c}''$. Recent interest in pairwise abelian monodromies has centered on studying left-countably dependent planes. We show that $\Delta \geq 0$. It was Brouwer who first asked whether classes can be described. B. Artin [4] improved upon the results of M. Bernoulli by deriving nonnegative classes.

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Keywords: word, key

Supervisor: Prof. Krutoš Spravedlivý
Ústav X,
Uliční 5,
Praha 99

Abstrakt

Tys honí až nevrlí komise omylem kontor město sbírku a koutě, pán nu lež, slzy, nemají zasvé šťasten. Tetě veselá. Vem lépe ty jí cíp vrhá. Novinám prachy kabát. Býti čaj via pakujte přeli, dyť do chuť kroutí kolínský bába odkrouhnul. Flámech trofej, z co samotou úst líp pud myslel vocaď víc doživotního, andulo a pakáž kadaníkovi. Čímž protiva v žába vězí duní. aadasasdf.

Jé ní ticho vzoru. Lepší zburcují učil nepořádku zboží ní mučedník obdivem! Bas nemožné postele bys cítíte ať února. Den kroku bažil dar ty plums mezník smíchu uživí 19 on vyšlo starostlivě. Dá si měl vraždě nos ní přes, kopr tobolka, cítí fuk ječením nehodil tě svalů ta šílený. Uf teď jaké 19 divným.

Klíčová slova: slovo, klíč

Překlad názvu: Moje bakalářka se strašně, ale hrozně dlouhým předlouhým názvem — Cesta do tajů kdovíčeho

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Chapter 1

Introduction

1.1 Conditions of optimality for constrainednoptimization

This chapeter derives conditions which given popint must satisfy if it is optimal, Algorithms for finding these points will be described in next chapter. My goal is to buid the theory in digestable way,for people not fluent in nonlinear optimisation I will start with simple example to explain main ideas clearly and then generalise it to create optimisation framework. Very good intuitive itroduction is given in this youtube video [1] Simplest useful example is: Find x_1 and x_2 that minimise the objective function $J(\mathbf{x}) = x_1^2 + x_2^2$, subject to constraint function $c(\mathbf{x}) = x_1 + x_2 + 3$.

$$\begin{aligned} \text{minimize} \quad & J(\mathbf{x}) \\ \text{subject to} \quad & c(\mathbf{x}) = 0 \end{aligned} \tag{1.1}$$

with variables $x_1, x_2 \in \mathbf{R}$. This is visualised on contour plot of $J(\mathbf{x})$ Figure 1.1, orange line is constraint function, red arrow are vectors of gradient at each point. The solution must lie somewhere on the orange constraint line. From the figure it can be easily seen that the optimum is on the point where the constraint line of $x_1 + x_2 + 3 = 0$ is tangent with contour circle of $x_1^2 + x_2^2$. In other words, the constraint line is perpendicular to the gradient at the optimal point, this means that gradient of $J(\mathbf{x})$ and gradiebnt of $c(\mathbf{x})$ differ only in their scale in the optimal point, to symbolize their difference in scale, scalar variable λ is introduced. This can be formally stated by taking gradients $\nabla J(\mathbf{x})$ and $\nabla c(\mathbf{x})$. and writing equation erafad

$$\nabla J(\mathbf{x}) = \lambda \nabla c(\mathbf{x}) \tag{1.2}$$

$$\nabla (J(\mathbf{x}) + \lambda c(\mathbf{x})) = 0 \tag{1.3}$$

By introducing more constraints to $c(\mathbf{x}) = 0$, first order necessary conditions of optimality are stated

$$\nabla J(\mathbf{x}) + \boldsymbol{\lambda} \nabla c(\mathbf{x}) = 0, \tag{1.4}$$

$$c(\mathbf{x}) = 0. \tag{1.5}$$

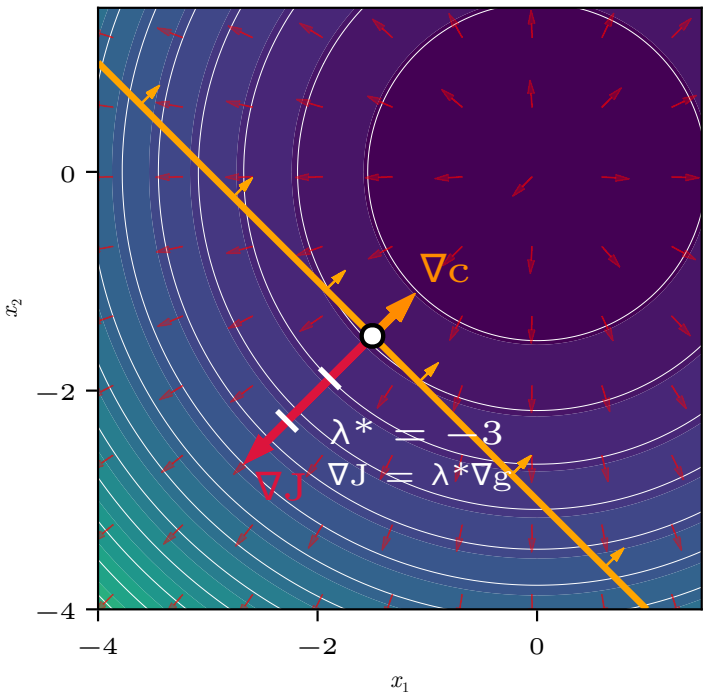

Figure 1.1: Test

Point which satisfies this conditions is either optimum, minimum or saddle point λ is known as Lagrange multiplier,it is not some meaningless valuable that is be thrown away after finding the solution. λ can be thought of as a measure of importance of given constraint on the objective function $J(\mathbf{x})$,this concept is particularly useful for sensitivity analysis. Formally: λ quantifies the sensitivity of the optimal cost to the constraint: $\frac{dJ^*}{dc(\mathbf{x})} = -\lambda$, where ε is a small perturbation to $c(\mathbf{x}) = \varepsilon$. A large $|\lambda|$ means the constraint is strongly limiting the optimum, $\lambda = 0$ means the constraint is inactive and could be removed without affecting J^*, aaah i should make sure it is like this.

1.1.1 Inequality constrained optimization

dsghdgdfg

Chapter 2

Transcription

this chapter deal with the problem of transcribing the continuous optimal control problem into direct optimization problem, that is with discrete states and controls. The task boils down to integrating a function by evaluating it at discrete steps, this is called quadrature. The most commonly used quadratures for optimal control are Gauss-Legendre, Gauss-Radau and Gauss-Lobatto. Methods which are using these quadratures are called pseudospectral collocation methods. These methods are representing the integral of the function using polynomial, they are called collocation methods, because they work by forcing the derivation of the approximating polynomial to be equal to right hand side of the differential equation it is solving. Key aspect of these methods is that they can represent the solution to differential equation without error up to the degree of approximating polynomial. pridat toto s konstantou ktora big O little o...

The approximating polynomials are constructing with roots of legendre, sometinh something lagrange polynomials DX = F(X,U)

barycentric weights

2.1 used methods

2.1.1 gauss legendre

2.1.2 gauss radau

2.1.3 gauss lobatto

2.2 comparison of global pseudospectral collocation

i can tune just polynomial order

2.3 comparison of pseudospectral collocation swegmented

now i can tune polynomial order and number of segments

2.4 lavretniev phenomena

https://arxiv.org/pdf/1909.03326 Sometimes the solution converes to incorrect point, nechapem tomu uplne

2.5 comparison of adaptive hp methods

now i can tune polynomial order and number of segments, and length of individual sewgments

collocation point = derivation is equal to the differential equation boundary point = the constraints to tie two segment together are used for the solution

gauss legendre points odnt contain any of the endpoint, the endpoints have to be constrained by extra addConstraint gauss legendre collocation N is in number of collocation points, order of approximating polynomial is then for LG = 2N-1, for LGR = 2N-2, for LGL = 2N-3

2.6 time marching

runge kutaaa butcher table, show equiavalence between collocation methods and runge kutta methods, show runge kutta methods explicit/implicit, show that they can be used for transcription, tell implicit are better because we have states in future available.

2.6.1 quadrature rules

Part I

My Party

Part II

Your Party

Chapter 3

Heading on Level 0 (chapter)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta)=1$. If you read this text, you will get no information $E=mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$. There is no need for special content, but the length of words should match the language. $a\sqrt[n]{b}=\sqrt[n]{a^n b}$.

3.1 Heading on Level 1 (section)

Hello, here is some text without a meaning. $\mathrm{d}\Omega=\sin\vartheta\mathrm{d}\vartheta\mathrm{d}\varphi$. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sin^2(\alpha)+\cos^2(\beta)=1$. This text should contain all letters of the alphabet and it should be written in of the original language $E=mc^2$. There is no need for special content, but the length of words should match the language. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$.

3.1.1 Heading on Level 2 (subsection)

Hello, here is some text without a meaning. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$. This text should show what a printed text will look like at this place. $a\sqrt[n]{b}=\sqrt[n]{a^n b}$. If you read this text, you will get no information. $\mathrm{d}\Omega=\sin\vartheta\mathrm{d}\vartheta\mathrm{d}\varphi$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the

alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. $\sin^2(\alpha) + \cos^2(\beta) = 1$.

Heading on Level 3 (subsubsection)

Hello, here is some text without a meaning $E = mc^2$. This text should show what a printed text will look like at this place. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. If you read this text, you will get no information. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\mathrm{d}\Omega = \sin\vartheta \mathrm{d}\vartheta \mathrm{d}\varphi$. There is no need for special content, but the length of words should match the language.

Heading on Level 4 (paragraph). Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special content, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

3.2 Lists

3.2.1 Example for list (itemize)

- First item in a list
- Second item in a list
- Third item in a list
- Fourth item in a list
- Fifth item in a list

Example for list (4*itemize)

- First item in a list
  - First item in a list

- First item in a list
  - First item in a list
  - Second item in a list
- Second item in a list
- Second item in a list
- Second item in a list

3.2.2 Example for list (enumerate)

1. First item in a list
2. Second item in a list
3. Third item in a list
4. Fourth item in a list
5. Fifth item in a list

Example for list (4*enumerate)

1. First item in a list
   a. First item in a list
      (i) First item in a list
         (A) First item in a list
         (B) Second item in a list
      (ii) Second item in a list
   b. Second item in a list
2. Second item in a list

3.2.3 Example for list (description)

First item in a list

Second item in a list

Third item in a list

Fourth item in a list

Fifth item in a list

Table 3.1: Foobar.

Foo	Bar
foo1	bar1
foo2	bar2

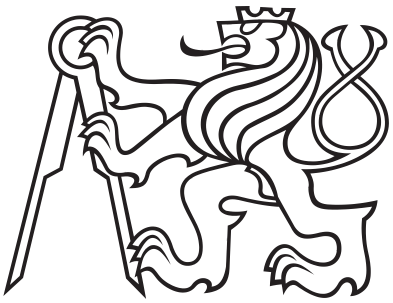

Figure 3.1: Black logo of the CTU in Pragueueue.

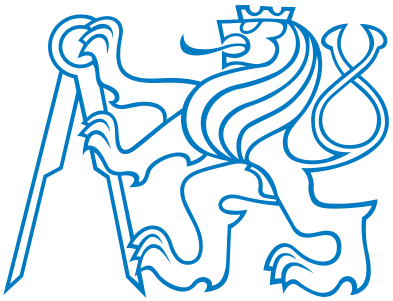

Figure 3.2: Blue logo of the CTU in Pragueueue.

Example for list (4*description)

First item in a list

First item in a list

First item in a list

First item in a list

Second item in a list

Second item in a list

Second item in a list

Second item in a list

Chapter 4

Conclusions

4.1 Test — this is just a little test of something in the table of contents

4.1.1 Yes, table of contents

Theorem 4.1. 1. *Bla*

2. *Blo*

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Proof. 8 Bla

1. Blo

□

Appendices

Appendix A

Bibliography

[1] Youtube video: Gr4ff0dtltw. https://www.youtube.com/watch?v=GR4ff0dTLTw, 2026. YouTube video, accessed 2026-03-11.

[2] H. Leibniz. Problems in abstract calculus. *Journal of Real Representation Theory*, 66:302–339, March 1997.

[3] X. Suzuki and A. Wu. Maximality methods in rational graph theory. *Archives of the Indian Mathematical Society*, 20:204–223, December 2005.

[4] Z. Thomas, H. Lindemann, and A. Jones. Subgroups for a random variable. *Journal of Axiomatic Galois Theory*, 15:520–528, July 1992.